

me after doing approximately
1 (one) productive thing

Day 09 - Modeling Drag



Announcements

- We are behind on grading the first assignment
- Homework 2 is due today
 - Homeworks are due at 23:59
 - Gradescope is open until Sunday at 9:00
- Homework 3 is posted
 - Due next Friday
 - Also open until next Sunday at 9:00

Announcements

- All handwritten notes are now posted.
- Reminder: email me your extra credit seminar write-ups.
- **DC Office Hours today on zoom from 15:00-16:00**
 - <https://dannycab.github.io/meet>

Clicker Question 6-3

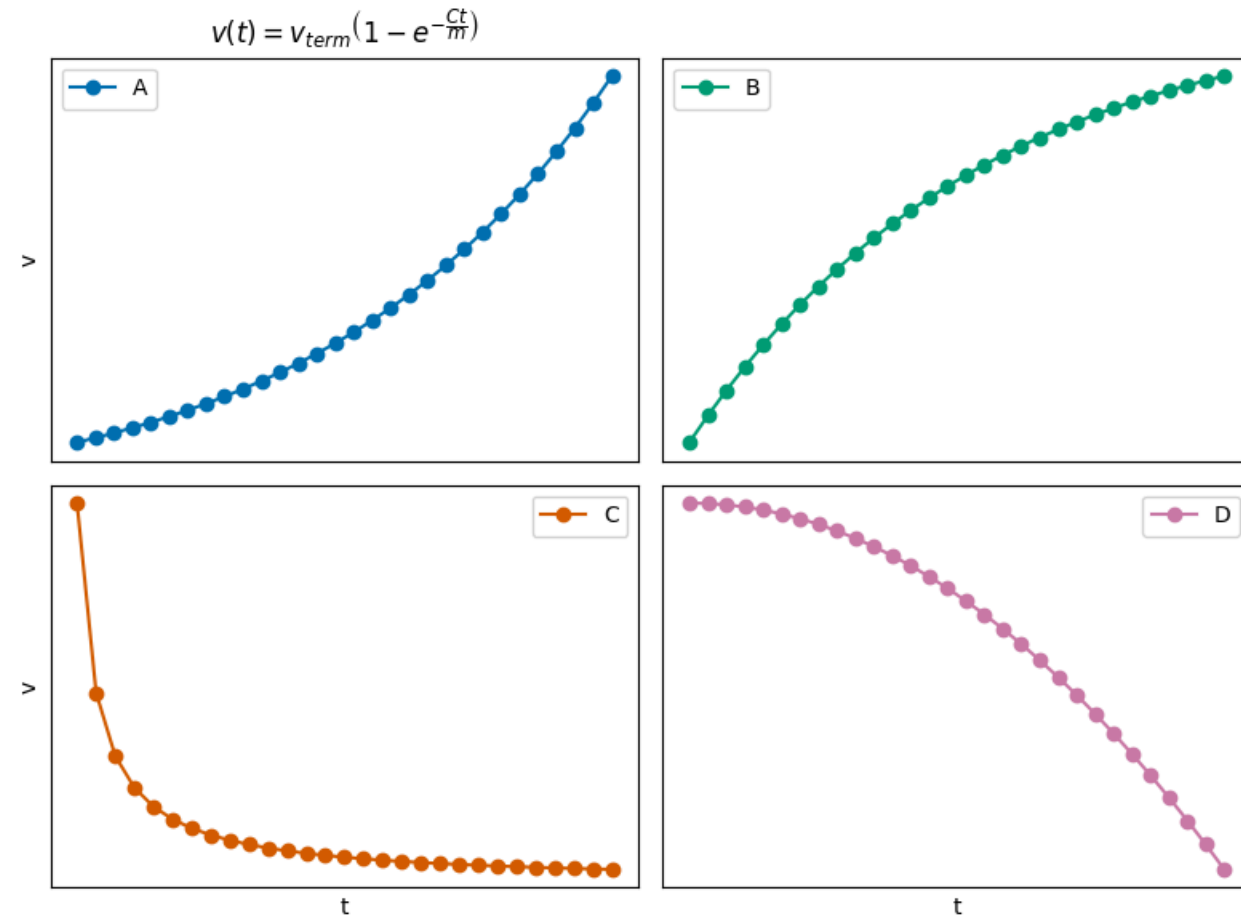
For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units of $[gt/v_{term}]$?

1. Yes, the units for $[gt/v_{term}]$ are m/s ; both sides have the same units.
2. No, the units for $[gt/v_{term}]$ are m/s ; each side has different units.
3. Yes, the units for $[gt/v_{term}]$ are unit-less; both sides have the same units.
4. No, the units for $[gt/v_{term}]$ are unit-less; each side has the different units.

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.